Workshop on Advanced Sampling Methodologies

Introduction to Sampling Design

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Workshop’s agenda

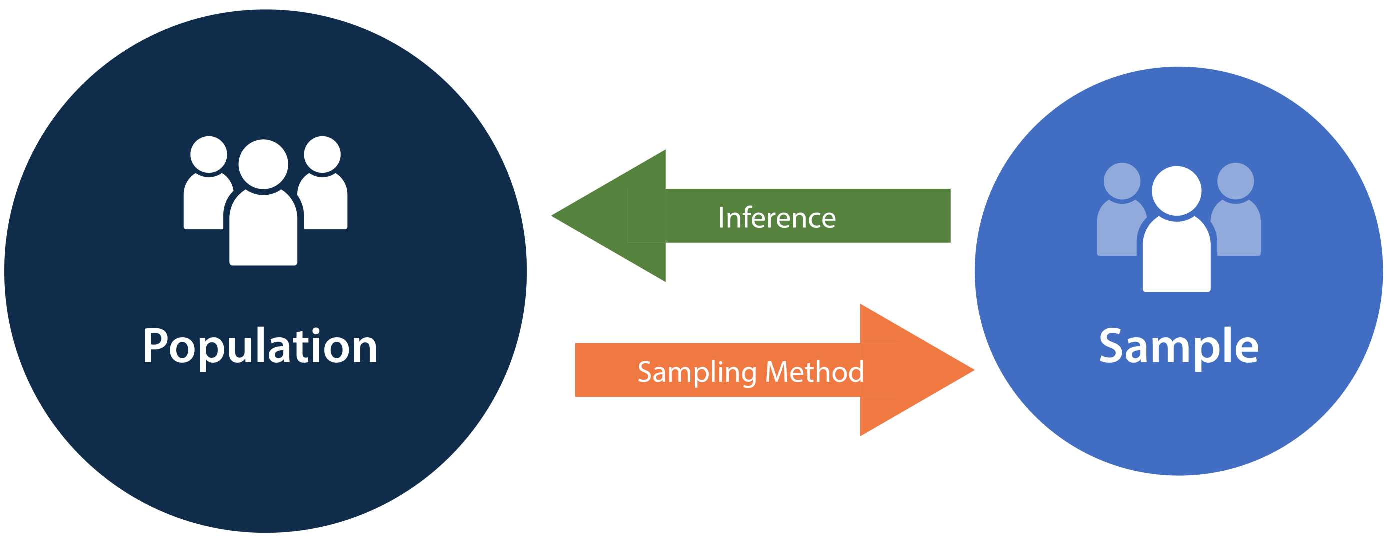
- **Day 1**: Introduction to QGIS
- **Day 2**: Automatic update of census EA and national sampling frame
- **Day 3**: Introduction to R programming
- **Day 4**: Introduction to Sampling Design
- **Day 5**: Advanced R programming
- **Day 6**: Earth observation (EO) for sampling design
- **Day 7**: AI and machine learning for sample selection
- **Day 8**: Simulation and optimization of sampling strategies
- **Day 9**: R Shiny for sampling design evaluation
- **Day 10**: Wrap-up and workshop evaluation

Today's agenda

- Introduction to sampling concepts
- Probability sampling methods
- Weighted sampling and unequal probability designs
- Practical implementation in R
- Design considerations and summary
- Practical exercises

Sampling

- **Definition**: sampling is the process of selecting a subset (called a **sample**) from a larger **population** to estimate characteristics of that population.
- **Purpose**: enables inference when measuring the entire population is costly or impractical. This offers the followwing advantages:
  - **Cost-effective**: sampling requires fewer resources than a full census.
  - **Feasibility**: some populations cannot be fully measured.
  - **Timely insights**: faster data collection for timely decision-making.



Key concepts of sampling (source: National Library of Medicine).

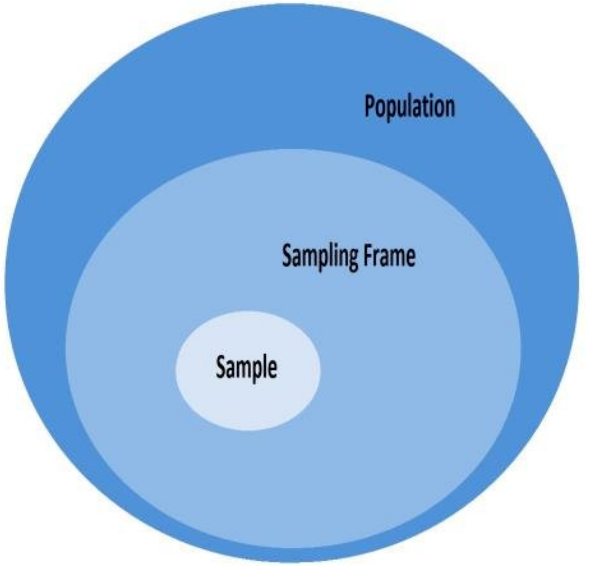
Key concepts

Population

- The **entire set of elements** (people, objects, events) you want to study.
- Provides the **scope** of inference for your study.
- **Example**: all university students in a country.

Sampling frame

- The **list or mechanism** used to identify and access members of the population.
- Ideally a sampling frame **should be**:
  - **Complete**: covers all members of population.
  - **Accurate**: no duplicates, no outdated entries.
  - **Accessible**: researcher can reach listed units.
- **Example**: an official student enrollment database.



Population, sampling frame, and sample (source: Ktistakis et al. 2021).

ℹ️ If the sampling frame does not fully represent the population, **sampling bias** may occur.

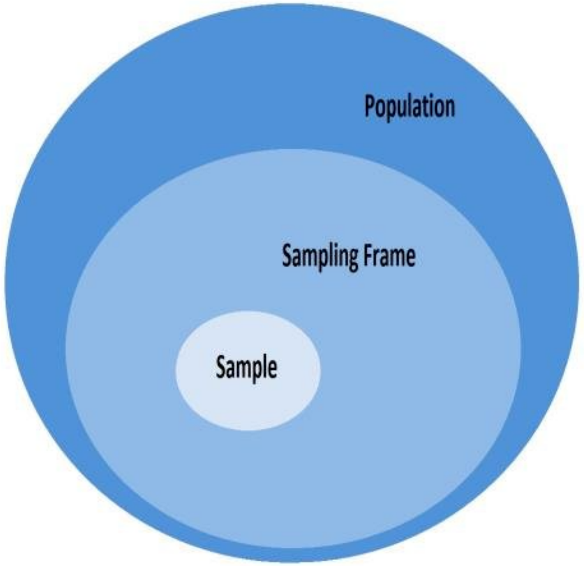
Key concepts

Sample

- A **subset of the population** selected for analysis.
- Used to make **inferences** about the population.
- **Example**: 1,000 students chosen from the national student population.

Sampling

- The **process or method** of selecting units from the population to form a sample.
- The **goal** is achieving a representative and unbiased sample.
- **Example**: students are selected randomly.

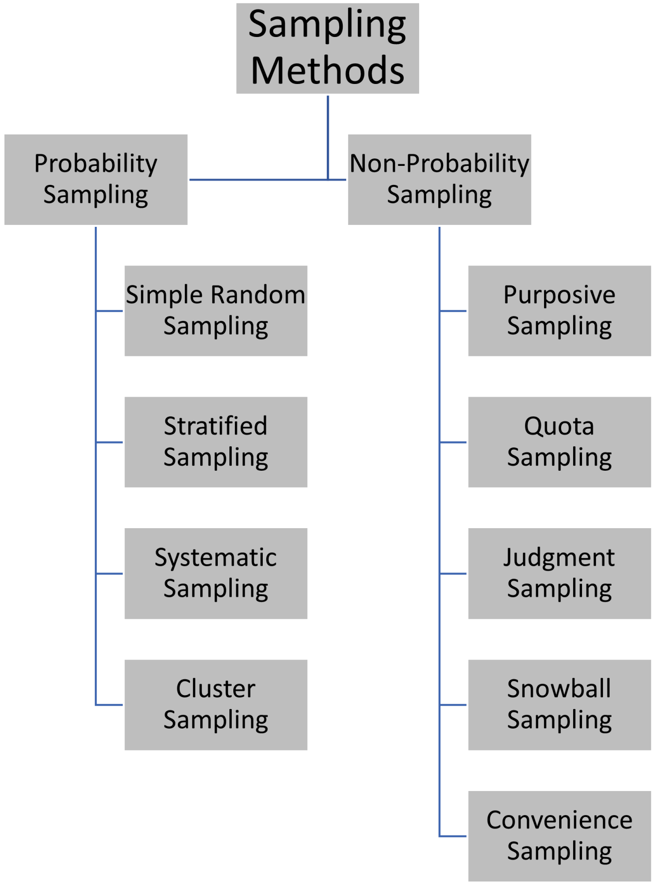


Population, sampling frame, and sample (source: Ktistakis et al. 2021).

ℹ️ **Sampling design** formalizes the plan for selecting a representative sample.

Sampling design

- A **sampling design** defines how samples are chosen and the probability each unit has of being selected.
- **Sampling design** includes aspects like:
  - **Probability versus non-probability sampling**
  - **Sampling technique**
  - **Sample size**
  - **Allocation strategies**
  - **Bias and representativeness**

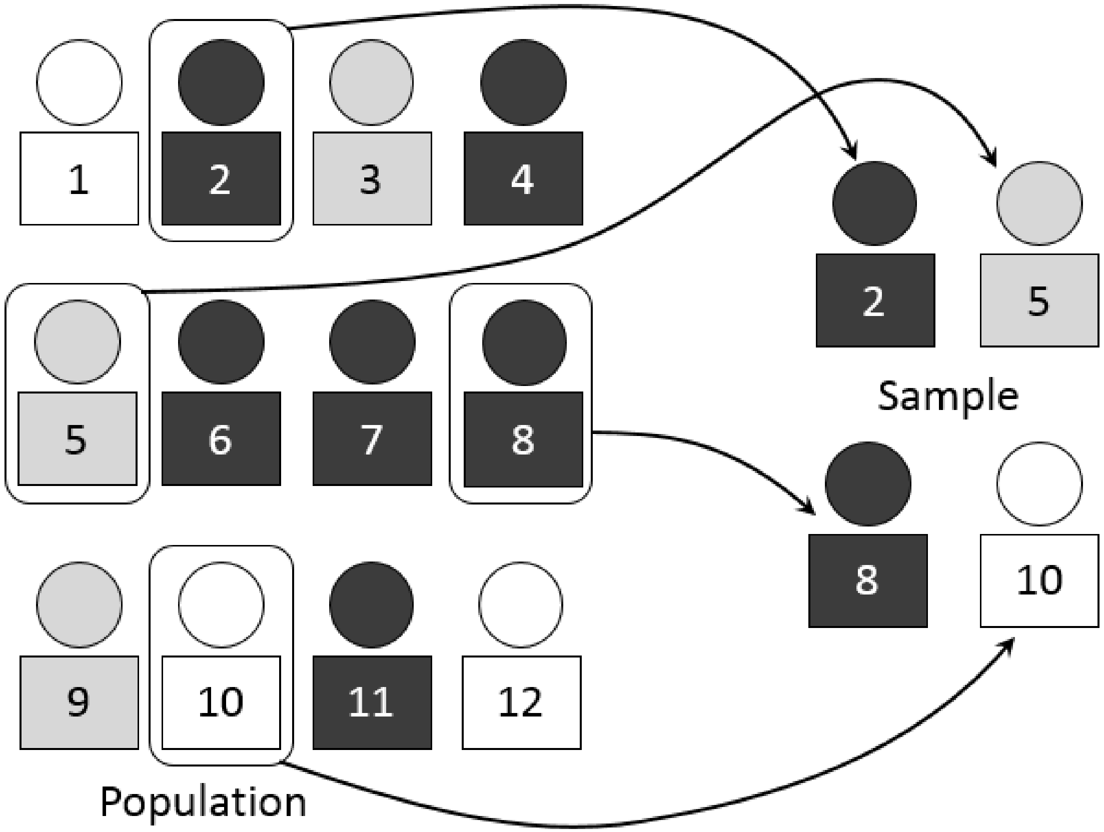


Sampling methods: probabilistic versus non-probabilistic.

ℹ️ In this workshop we will cover **probability sampling** only.

Simple random sampling

- Every possible subset of size *n* from a population *N* has an **equal probability of being selected**.
- **Used when** complete list of the population is available.



Simple random sampling (source: Wikipedia)

Simple random sampling

Sample mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Sample variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

Population mean

$$\mu = \frac{1}{N} \sum_{i=1}^{N} x_i$$

Population variance

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu)^2$$

x: sampled values

μ: population mean

σ^2: population variance

n: sample size

N: population size

ℹ️ The sample mean can be computed by substituting N with n, while the sample variance by substituting N with n and $n - 1$ in the denominator.

Simple random sampling

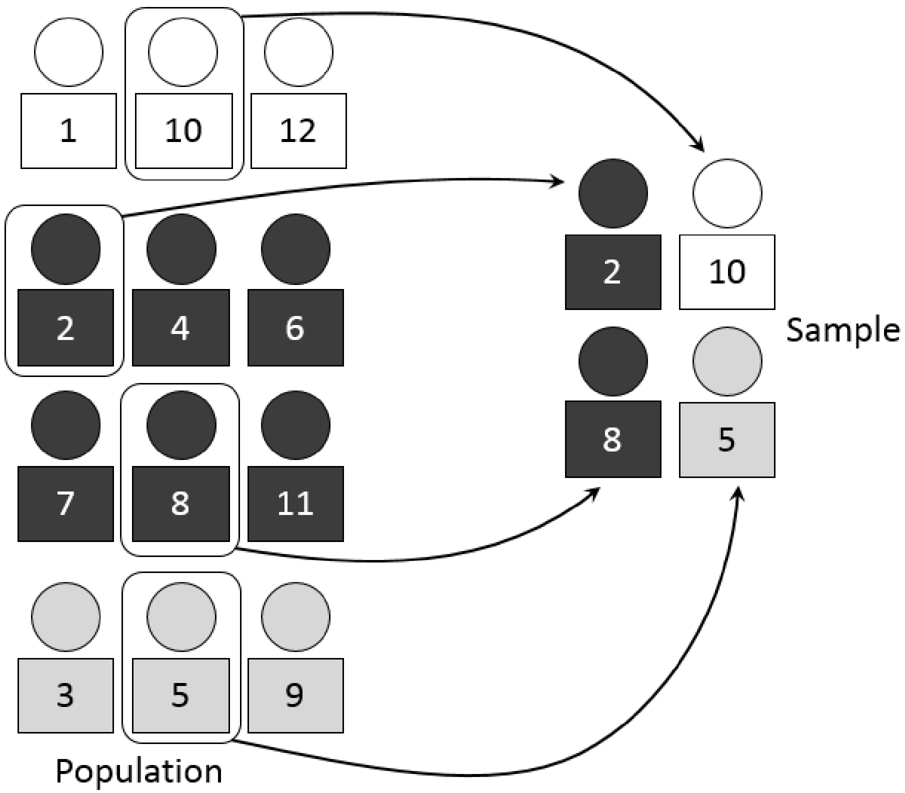
Implementation in R

```r
library(tidyverse)                              # Load tidyverse for data manipulati
set.seed(123)                                   # Set seed for reproducibility

N <- 10000                                      # Define population size
pop <- data.frame(id = 1:N,                     # Create population dataframe with I
                  y  = rlnorm(N,                # Generate log-normal random values
                  meanlog = 2,
                  sdlog = 0.5))

n <- 500                                        # Define sample size
s_ids <- sample(pop$id, size = n,               # Randomly select n IDs without repl
                replace = FALSE)
samp <- pop |> filter(pop$id %in% s_ids) # Extract sampled observations

ybar_hat <- mean(samp$y)                        # Sample mean of y
Y_hat <- N * ybar_hat                           # Estimate of population total (N *
```

ℹ️ Try to run the code in your Script Editor.

Stratified sampling

- Population is divided into **homogeneous subgroups (strata)** and a **simple random sample** is drawn **within each stratum**.
- Improves efficiency by reducing variability **within strata**.



Stratified sampling (source: Wikipedia)

Stratified sampling

Sample mean:

$$\bar{x}_{st} = \sum_{h=1}^{H} W_h \bar{x}_h$$

Approximate variance

$$\text{Var}(\bar{x}_{st}) = \sum_{h=1}^{H} W_h'^2 \frac{\sigma_h'^2}{n_h}\left(1 - \frac{n_h}{N_h}\right)$$

Population mean

$$\mu = \frac{1}{N} \sum_{h=1}^{H} \sum_{i=1}^{N_h} x_{hi}$$

Population variance

$$\sigma^2 = \frac{1}{N} \sum_{h=1}^{H} \sum_{i=1}^{N_h} (x_{hi} - \mu)^2$$

N_h: size of stratum h;

n_h: sample size for stratum h;

N: total population size;

$W_h = \frac{N_h}{N}$: population weight of h,

$f_h = \frac{n_h}{N_h}$: sampling fraction in h,

$\bar{x}_h$: sample mean in h,

$s_h'^2$ =: sample variance in h.

Stratified sampling

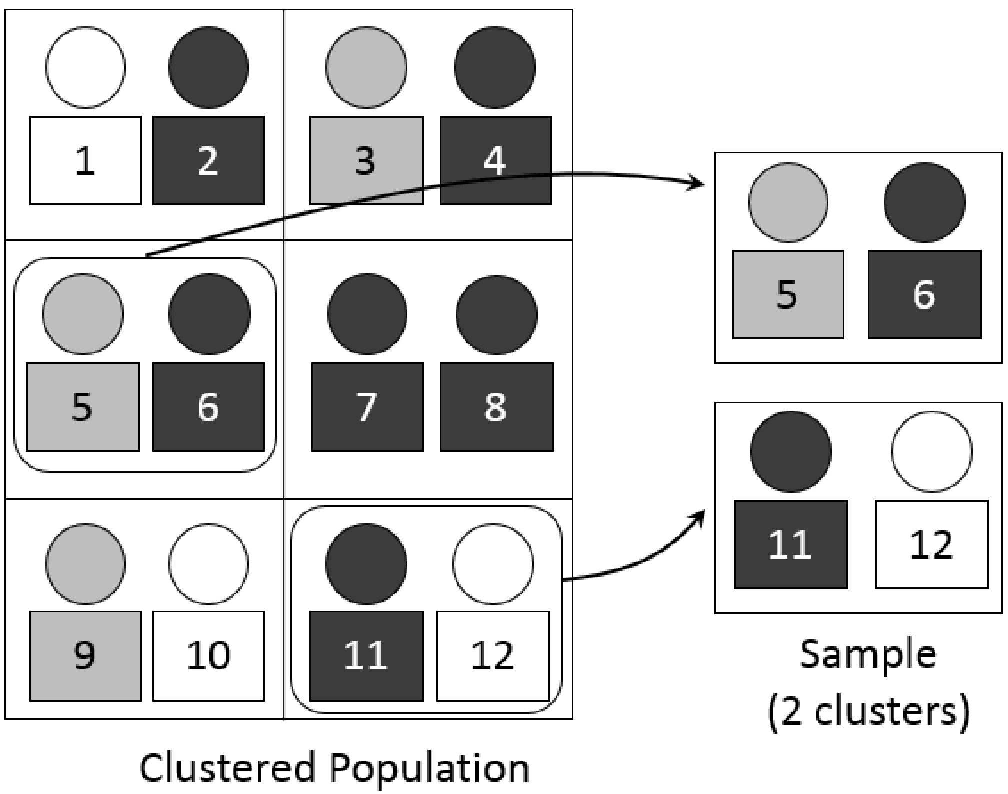
Implementation in R

```r
set.seed(7)                                # Set seed for reproducibility
H <- 3                                     # Number of strata
N_h <- c(4000, 3000, 3000)                 # Population sizes for each stra

stratum <- rep(1:H, times = N_h)           # Vector labeling each unit's st
pop <- data.frame(id = 1:sum(N_h),         # Build full population dataset
                  h = stratum,             #   - stratum membership
                  y = c(rnorm(N_h[1], 50, 10),  #   - outcomes for stratum
                        rnorm(N_h[2], 60, 15),  #   - outcomes for stratum
                        rnorm(N_h[3], 55, 12))) #   - outcomes for stratum

N <- nrow(pop)                             # Total population size
W_h <- N_h / N                             # Stratum weights (proportion of
n <- 600                                   # Desired total sample size

n_h <- round(n * W_h)                      # Allocate sample size per strat
```

ℹ️ Try to run the code in your Script Editor.

Cluster sampling

- Population is divided into **clusters**, often based on geography or natural groupings.
- A **random sample of clusters** is selected, and **all (or a sample of) units within selected clusters** are observed.



Cluster sampling (source: Wikipedia)

Cluster sampling

Cluster sample mean

$$\bar{x}_{cl} = \frac{1}{n_c} \sum_{c \in S} \bar{x}_c$$

Approximate variance of $\bar{x}_{CL}$:

$$\mathrm{Var}(\bar{x}_{cl}) = \frac{1}{n_c}\left(1 - \frac{n_c}{C}\right) S_c^2$$

Population mean

$$\mu = \frac{1}{N} \sum_{c=1}^{C} \sum_{i=1}^{N_c} x_{ci}$$

Population variance

$$\sigma^2 = \frac{1}{N} \sum_{c=1}^{C} \sum_{i=1}^{N_c} (x_{ci} - \mu)^2$$

N: Population size
C: Number of clusters in the population
N_c: Size of cluster
n_c: Number of clusters sampled
x_{ci}: Value of the i-th element in cluster c
$\bar{x}_c$: Mean of cluster c
$\bar{x}_{cl}$: Mean of sampled clusters
μ: Population mean
σ^2: Population variance
S_c^2: Variance of cluster means
S: Set of sampled clusters

Cluster sampling

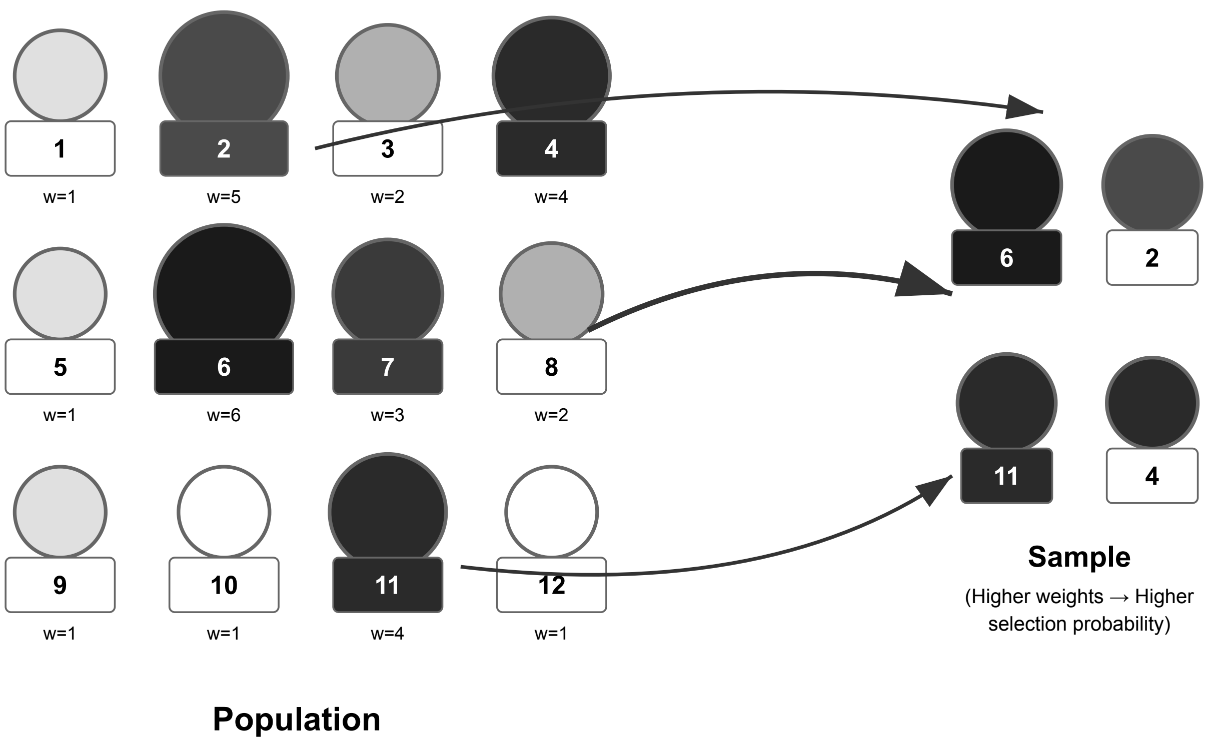
Implementation in R

```
set.seed(11)                                  # Set random seed for reproducibilit
J <- 100                                      # Number of clusters in the populati
mbar <- 20                                    # Average cluster size
rho <- 0.15                                   # Intra-cluster correlation (ICC)

M_j <- rpois(J, mbar-1) + 1                   # Cluster sizes (Poisson around mbar
mu <- 50                                      # Overall population mean
sigma2 <- 100                                 # Total variance of outcomes
tau2 <- rho * sigma2                          # Between-cluster variance component
sigma2_within <- (1-rho) * sigma2             # Within-cluster variance component
cluster_effect <- rnorm(J, 0, sqrt(tau2)) # Random cluster effects ~ N(0, t

data <- do.call(rbind, lapply(1:J, function(j){   # Generate population dat
  m <- M_j[j]                                 # Size of cluster j
  y <- mu + cluster_effect[j] +               # Mean + cluster effect +
       rnorm(m, 0, sqrt(sigma2_within))# individual error within cluster
```

ℹ️ Try to run the code in your Script Editor.

Weighted sampling

- Each unit in the population is assigned a **weight** reflecting its **probability of selection**.
- Units with higher weights have **greater influence** on the estimate.
- Often used in **complex surveys** or **unequal probability sampling** designs.



Probability weighted sampling.

Weighted sampling

Population mean (weighted)

$$\mu = \frac{\sum_{i=1}^{N} w_i x_i}{\sum_{i=1}^{N} w_i}$$

Population variance (weighted)

$$\sigma^2 = \frac{\sum_{i=1}^{N} w_i (x_i - \mu)^2}{\sum_{i=1}^{N} w_i}$$

Weighted sample mean

$$\bar{x}_w = \frac{\sum_{i \in S} w_i x_i}{\sum_{i \in S} w_i}$$

Approximate variance of $\bar{x}_w$

$$\mathrm{Var}(\bar{x}_w) \approx \sum_{i \in S} \frac{w_i^2}{(\sum_{i \in S} w_i)^2} (x_i - \bar{x}_w)^2$$

N: Population size
x_i: Value of the i-th element in the population
w_i:Weight of element i (proportional to size)
S: Set of sampled elements
$\bar{x}_w$: Weighted sample mean
μ: Population mean (weighted)
σ^2: Population variance (weighted)
$\mathrm{Var}(\bar{x}_w)$: Approximate variance of weighted sample mean

Weighted sampling

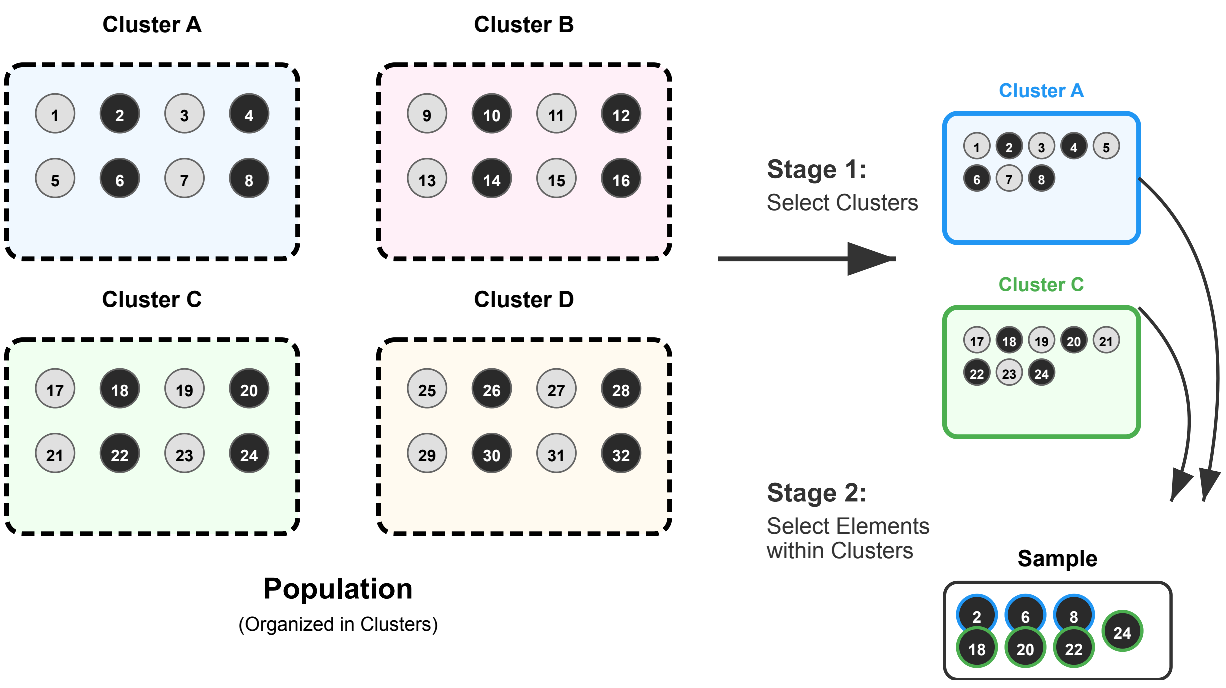
Implementation in R

```r
set.seed(123)                                # Fix seed for reproducibility

N <- 1000                                    # Population size
x <- runif(N, 1, 10)                         # Size measure
y <- 5 + 2*x + rnorm(N)                      # Outcome correlated with x
pop <- data.frame(id=1:N, x=x, y=y)          # Store in a dataframe

n <- 100                                     # Sample size
srs <- pop[sample(1:N, n), ]                 # Draw SRS without replacement
mean_srs <- mean(srs$y)                      # Sample mean (unweighted)
mean(pop$y)                                  # True population mean (for comp

prob <- x / sum(x)                           # Selection probs proportional t
samp_ids <- sample(1:N, n, prob=prob)        # Draw weighted sample
pps <- pop[samp_ids, ]                       # Extract sampled units
pps$w <- 1/prob[samp_ids]                    # Compute weights = 1 / inclusio
```

ℹ️ Try to run the code in your Script Editor.

Multistage sampling

- Sampling is conducted in **two or more stages**, selecting units at each stage.
- Common design:
  - Stage 1: Randomly sample **clusters** (e.g., schools).
  - Stage 2: Randomly sample **elements within clusters** (e.g., students).
- Useful when a complete list of all population units is unavailable.



Multistage sampling.

Multistage sampling

Population mean

$$\mu = \frac{1}{N}\sum_{c=1}^{C}\sum_{i=1}^{N_c} x_{ci}$$

Population variance

$$\sigma^2 = \frac{1}{N}\sum_{c=1}^{C}\sum_{i=1}^{N_c}(x_{ci} - \mu)^2$$

2-stage sample mean:

$$\bar{x}_{ms} = \frac{1}{n_c}\sum_{c \in S}\bar{x}_c, \quad where \;\; \bar{x}_c = \frac{1}{n_{ci}}\sum_{i \in S_c} x_{ci}$$

Approximate variance of $\bar{x}_{ms}$

$$\mathrm{Var}(\bar{x}_{ms}) \approx \frac{1}{n_c}\left(1 - \frac{n_c}{C}\right) S_c^2 + \frac{1}{n_c^2}\sum_{c \in S}\frac{S_{ci}^2}{n_{ci}}$$

N Total population size

C: Number of clusters in population

N_c: Size of cluster c

n_c: Number of clusters sampled

S: Set of sampled clusters

x_{ci}: Value of element i in c

S_c: Variance of cluster means in population

S_{ci}^2: Variance of elements within sampled c

n_{ci}: Number of elements sampled from c

$\bar{x}_c$: Mean of sampled elements in c

Multistage sampling

Implementation in R

```r
set.seed(42)  # Ensure reproducibility

# --- Stage 0: Define population ---
J <- 50                        # Number of primary clusters (e.g., schools)
mbar <- 30                     # Average number of units per cluster (e.g., studen
mu <- 60                       # Overall population mean
sigma2 <- 100                  # Total variance

# Generate cluster sizes (first-stage)
M_j <- rpois(J, mbar-1) + 1   # Random cluster sizes around mbar

# Generate cluster-level effects
tau2 <- 0.2 * sigma2             # Between-cluster variance
sigma2_within <- sigma2 - tau2 # Within-cluster variance
cluster_effect <- rnorm(J, 0, sqrt(tau2))  # Random cluster effects
```

ℹ️ Try to run the code in your Script Editor.

Summary table

Sampling Method	Key Characteristics	Strengths	Limitations
Simple Random	Every unit has an equal chance; requires full sampling frame	Unbiased; easy to implement	Inefficient for large or dispersed populations; ignores subgroups
Stratified	Population divided into homogeneous strata; random sample within each stratum	Higher precision; ensures subgroup representation	Requires stratum info for all units; complex analysis
Cluster	Population divided into clusters; some clusters randomly selected	Cost-effective; useful for large/geographic populations	Higher variance if clusters are homogeneous
Weighted	Units sampled with unequal probabilities; weights used in estimation	Adjusts for unequal probabilities; enables oversampling	Requires accurate weights; complex variance estimation
Multistage	Sampling occurs in two or more stages (e.g., clusters and elements)	Logistically efficient; scalable	Can increase variance; complex design and analysis
Non-Probability	Units selected based on convenience or judgment,	Easy, fast, cheap	Cannot generalize to population; prone to

Sampling design

Key considerations

1. Define population and objectives
2. Establish reliable sampling frame
3. Choose method aligned with objectives and resources
4. Calculate appropriate sample size
5. Minimize bias & sampling error
6. Document design and limitations thoroughly

Take home

- Sampling allows inference from a subset when a full census is impractical.
- Probability sampling ensures representativeness and enables unbiased estimates.
- Stratification and weighting improve precision and adjust for unequal probabilities.
- Cluster and multistage sampling enhance feasibility for large or dispersed populations.
- Careful design, sample size calculation, and bias minimization are essential for reliable results.

References

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R Core Team. (2025). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org

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Exercise

Please download the R script with exercises from GitHub and try to complete it.